

Forest understory vegetation study: current status and future trends

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Abstract

Understory vegetation accounts for a large proportion of floral diversity. It provides various ecosystem functions and services, such as productivity, nutrient cycling, organic matter decomposition and ecosystem self-regeneration. This review summarizes the available literature on the current status and progress of the ten most studied branches of understory vegetation on both its structural and functional aspects based on global climate change and forest management practices. Future research directions and priorities for each branch is suggested, where understory vegetation in response to the interplay of multiple environmental factors and its long-term monitoring using ground-based surveys combined with more efficient modern techniques is highlighted, although the critical role of understory vegetation in ecosystem processes individually verified in the context of management practices or climate changes have been extensively investigated. In summary, this review provides insights into the effective management of the regeneration and restoration of forest ecosystems, as well as the maintenance of ecosystem multilevel structures, spatial patterns, and ecological functions.

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Introduction

Understory vegetation is a vital stratum of forests, including seedlings (< 1.5 m tall), shrubs, herbs, bryoids (i.e., mosses and lichens), and lianas. Both vascular (woody and non-woody) and nonvascular (liverworts, hornworts, and mosses) plants are essential components of understory vegetation^[1], which plays important roles in forest ecosystem structure and functioning (e.g., biodiversity, regeneration, biomass, and nutrient content, as well as functional traits) and provision of ecological services (e.g., soil nutrient cycling and biological processes, water conservation, as well as mitigating emissions of greenhouse gas from carbon (C) sequestration and nitrogen (N) dynamics) (Fig. 1)^[2,3]. They can also provide food, shelter, and habitat for animal species, especially for soil arthropods and large herbivores^[4]. Nevertheless, the composition and distribution of understory vegetation are strongly influenced by competition (generalizability) or facilitation (specificity) from the overstory trees^[5–7]. Understory species are always seen as 'noxious

weeds' with exploitative competition for light, water, and nutrients or interference competition involving allelopathic effects against target overstory trees, especially plantation forests^[8,9]. Moreover, overstory management, mainly thinning and pruning practices, can directly and indirectly affect the formation of understory vegetation, which in turn regulates understory community resistance, adaptation, and resilience to general climate changes (e.g., drought, global warming, and N deposition), extreme climate events (e.g., fire, heatwaves, and freezing rain), and other adversities (e.g., strong irradiance, pests, and pathogens)^[10–12]. As a result, more experiments based on understory removal and overstory management are being carried out to explore the role of understory vegetation in ecosystem function, with some controversial results^[9,13,14]. This review recapitulated the current status, progress, and future direction of forest understory vegetation studies. We expect that this review can help understand the mechanisms of understory developments to support ecosystem multifunctionality and sustainability against unpredictable global climate